

SMOKING AND PERIODONTAL DISEASE

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OUTLINE

- Introduction
- Classification of smokers
- Smoking and periodontal disease
- Effect of smoking on Dental structure
- Smoking and Periodontal therapy
- Smoking cessation

INTRODUCTION

- Periodontitis risk factors include genetics, systemic diseases, and local plaque biofilm retention.
- Smoking is a major risk factor for periodontitis.
- Smoking is a public health threat in both upper and lower-income countries.
- Global daily tobacco smoking prevalence rates have decreased, but with population growth, the number of daily smokers has increased to 1.1 billion in 2015.

CONT'D

- Highest prevalence rates are in high-income and upper-middle-income countries, with lower rates in lower-middle-income and low-income countries.
- WHO has set a 30% relative reduction target between 2010 and 2025, with the Americas expected to attain this.

Smoking's Health Impact

- Harmful to all body organs and associated with various diseases.
- Causes lung cancer, heart disease, stroke, emphysema, bronchitis, and cancers of various organs.
- Half of long-term smokers die early, with an average loss of 20 years before 70.
- Major causes include lung cancer, chronic obstructive pulmonary disease, and coronary heart disease.

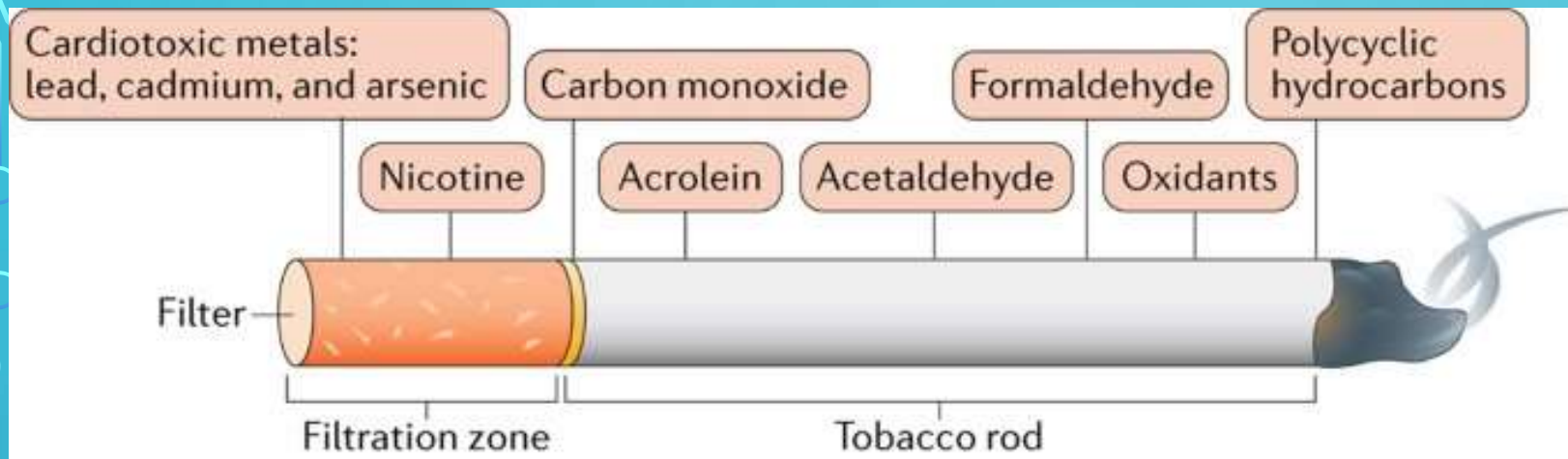
A Global Perspective

- As many as 6.25 trillion **cigarettes** are smoked annually across the globe.⁹⁵
- Tobacco kills 6 million people each year.¹⁴²
- Only 30% of the world's population have access to **smoking cessation** services.¹⁴⁴
- Only 22% of the world's population are protected by smoke-free laws.¹⁴⁴

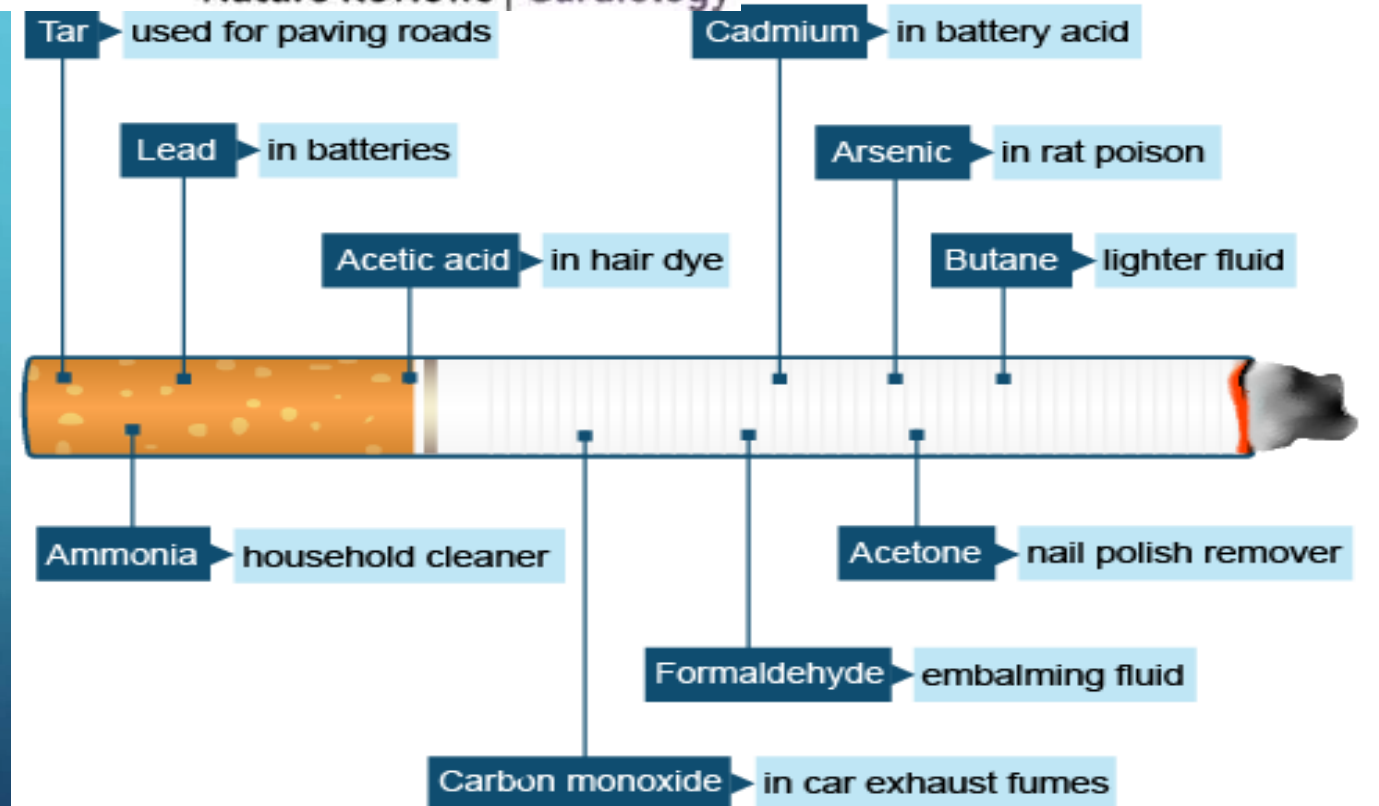


CHEMICAL COMPOSITION

- Tobacco smoke consists of a gaseous phase and a solid phase.
- **Gas phase:** CO, NH₃, formaldehyde, hydrogen cyanide, and over 60 carcinogens.
- **Particulate phase:** Nicotine, "tar," benzene, and benzo(a)pyrene.
- **Tar:** an inhaled sticky brown substance, stains fingers and teeth.
- **Nicotine:** an alkaloid, evaporates when cigarette is lit and is quickly absorbed in the lungs.
- Highly addictive, it causes blood pressure rise, increased heart and respiratory rates, and peripheral vasoconstriction.



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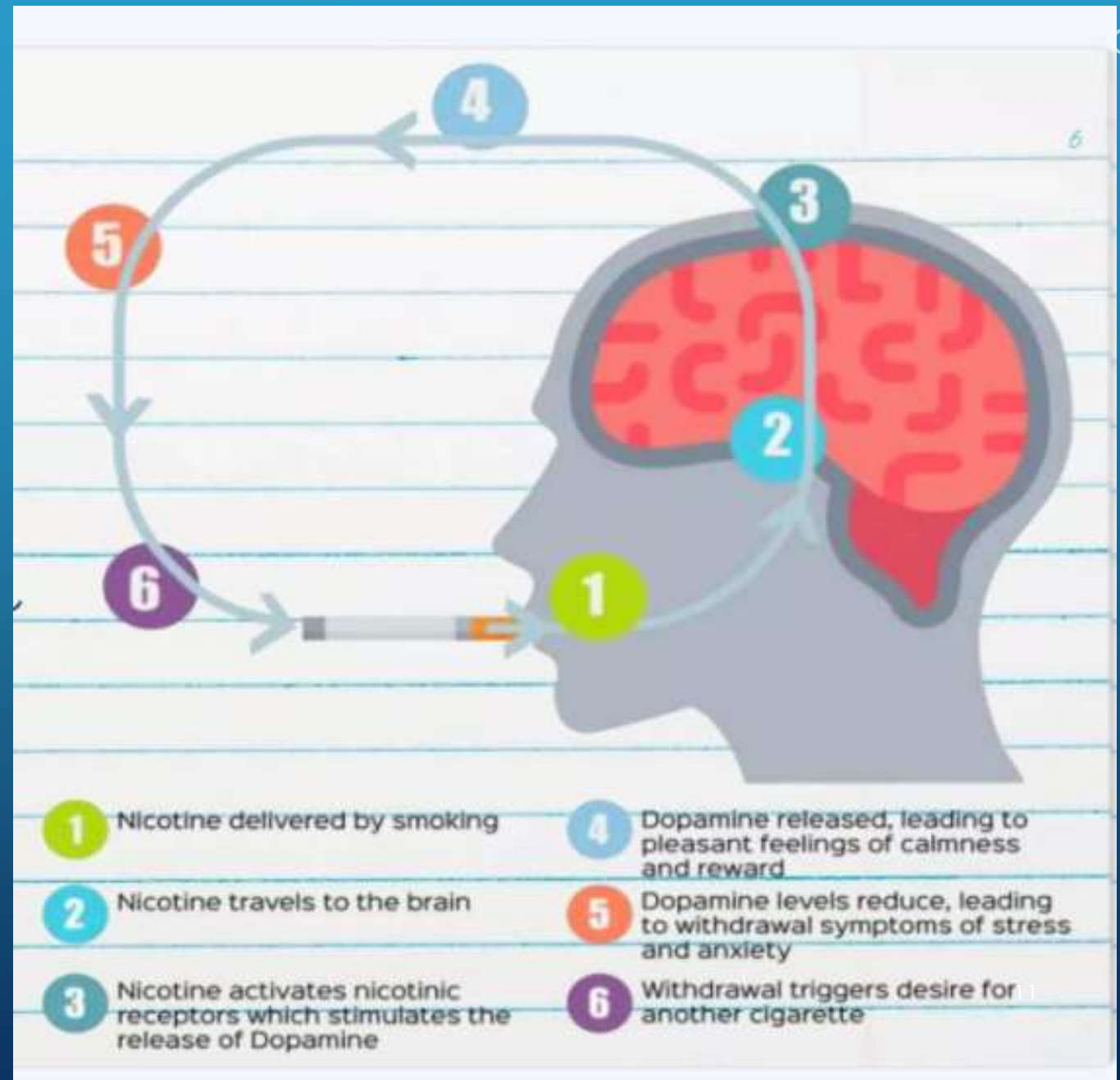




NICOTINE

- Main alkaloid found in tobacco
- Responsible for addictive potential
- Most active compound in smoke
- During smoking- inhaled nicotine is absorbed through the **pulmonary venous** rather than systemic venous system- reaches the brain in 10-20 secs.
- Acts on Nicotine acetylcholine receptors , on binding increases level of NT including dopamine in the reward circuit
- Euphoria and addiction

NICOTINE



Smokeless Tobacco

- Available in various forms: ground/grated, "moist snuff," dry snuff, and chewed tobacco.
- Delivered as a small tea-bag-like sachet under the lip.
- Ground to a powder for inhalation into the nasal cavity.
- Can be coarsely cut (chewing tobacco) and placed inside the cheek

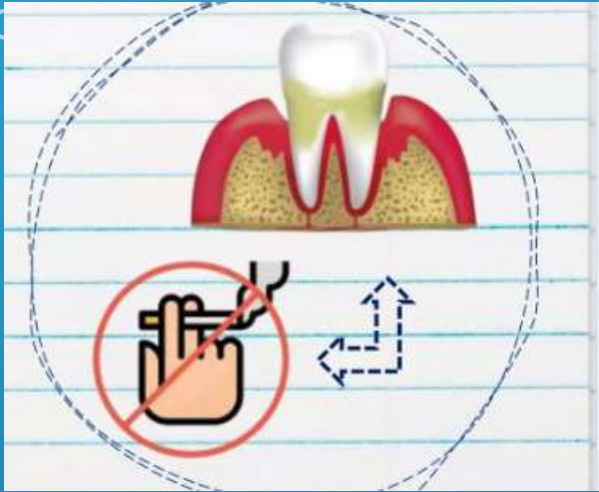
Classification of Smokers

- According to Centers for Disease Control (CDC)
 - ✓ **Current smokers:** those who had smoked 100 or more cigarettes over lifetime
 - ✓ **Former Smokers:** had smoked 100 or more cigarettes in their lifetime but were not currently smoking
 - ✓ **Passive Smokers:** had not smoked 100 or more cigarettes in their lifetime

Pack years = Number of packs smoked per day ×
Number of years of smoking

CONT'D

- According to number of cigarettes smoked/day
 - ✓ **Heavy smokers:** those who smoked more than 20 cigarettes/day
 - ✓ **Light smokers:** those who smoked less than 19 cigarettes/ day
- Current smoking status is the minimum, but cumulative exposure is more important.



SMOKING AND PERIODONTAL DISEASE

- Smoking is a major risk factor for periodontal destruction, with pocket depth, attachment loss, and alveolar bone loss more prevalent in smokers.
- Tobacco smoking significantly influences the development and progression of periodontitis in a dose-dependent manner
- Former smokers were 1.7 times more likely to have periodontitis than those who had never smoked.



SMOKING AND PERIODONTAL DISEASE

Table 14.2: Impacts of smoking on periodontal diseases

Periodontal disease	Impact of smoking
Gingivitis	• Decreased gingival inflammation and bleeding on probing
Periodontitis	• Increased prevalence and severity of periodontal destruction • Increased pocket depth, attachment loss and bone loss • Increased rate of periodontal destruction • Increased prevalence of severe periodontitis • Increased tooth loss • Increased prevalence with increased number of cigarettes smoked per day • Decreased prevalence and severity with smoking cessation; significantly more prevalent in current smokers than in non-smokers and former smokers

CONT'D

Immunology

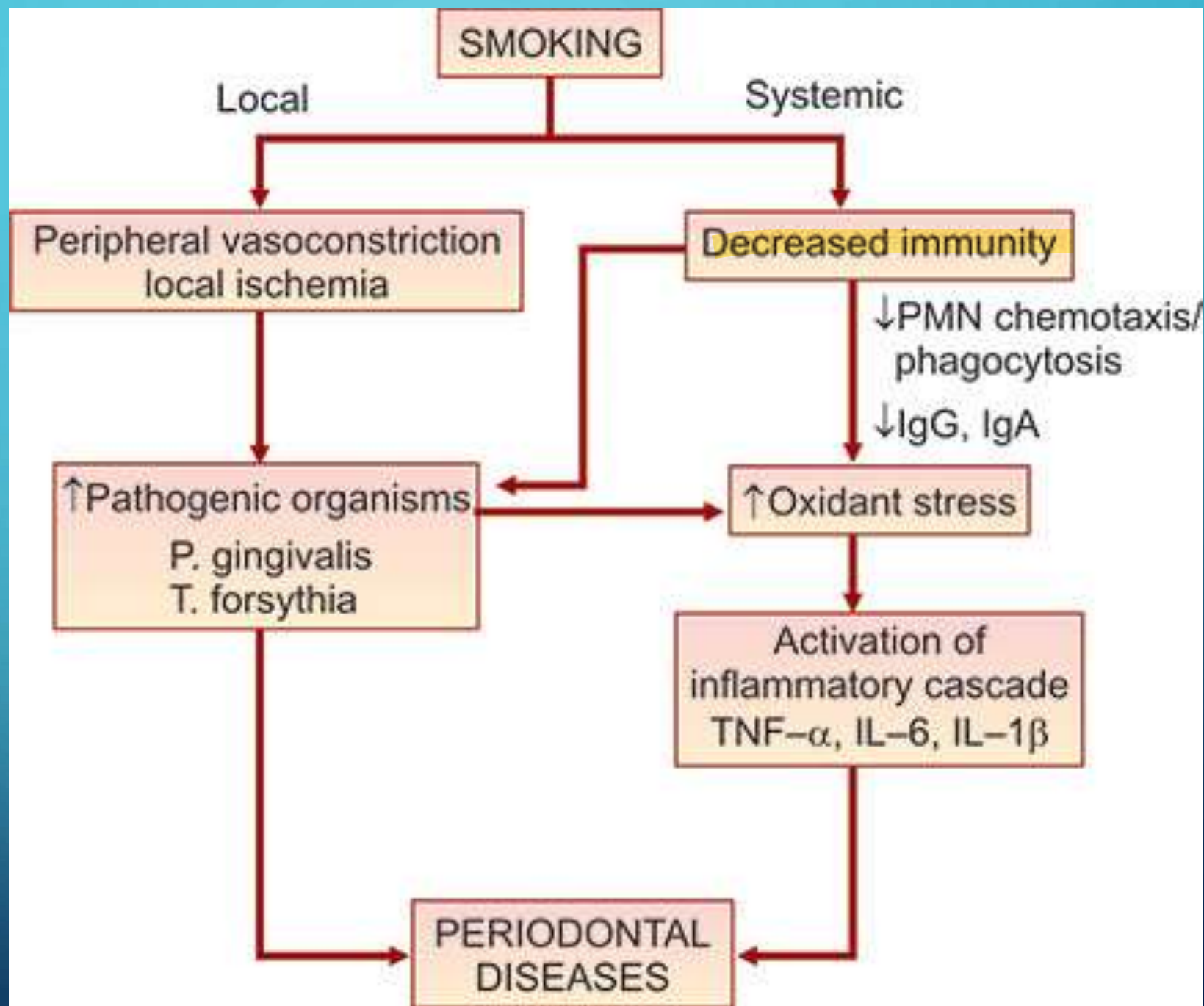
1. Altered neutrophil chemotaxis, phagocytosis and oxidative burst.
2. Increased tumor necrosis factor-alpha (TNF- α) and prostaglandin E₂ (PGE₂) in gingival crevicular fluid (GCF).
3. Increased production of PGE₂ by monocyte in response to lipopolysaccharides (LPS).
4. Immunoglobulin G2 (IgG₂) level is reduced suggesting reduced protection against periodontal infection.
5. Nicotine, a major component of tobacco, adversely affects fibroblast function. Nicotine levels in GCF can be up to nearly 300 times that of nicotine plasma concentrations in smokers.
6. Nicotine suppresses osteoblast proliferation, while stimulating alkaline phosphatase activity. Products of nicotine form a layer on root surface, which can alter fibroblast attachment, and hence decreased collagen production.
7. Tobacco products alter normal reparative and regenerative potential of periodontium.

CONT'D

Table 14.3: Effects of smoking on the etiology and pathogenesis of periodontal disease

<i>Factor</i>	<i>Impact of smoking</i>
Microbiology	• No effect on rate of plaque accumulation • Increased colonization of shallow periodontal pockets by periodontal pathogens • Increased levels of periodontal pathogens in deep periodontal pockets
Immune-inflammatory response	• Altered neutrophil chemotaxis, phagocytosis and oxidative burst • Increased tumor necrosis factor-α and Prostaglandin E2 in GCF • Increased neutrophil collagenase and elastase in GCF
Physiology	• Decreased gingival blood vessels with increased inflammation • Decreased GCF flow and bleeding on probing with increased inflammation • Decreased subgingival temperature • Increased time needed to recover from local anesthesia

CONT'D





Effect of Smoking on Dental structure

- **TEETH:** Black or brown stains on tooth surfaces, increased calculus formation, and poor oral hygiene.
- **GINGIVA:** Grayish discoloration and hyperkeratosis of the gingiva, changes in gingival epithelium.
- **Gingival blood vessels:** Smokers have higher proportion of small blood vessels and lower proportion of large blood vessels, possibly suppressing inflammatory response.

CONT'D

- Possible relationship between smoking and acute necrotizing ulcerative gingivitis (ANUG) due to;
 - ✓ Vasoconstriction of gingival blood vessels
 - ✓ Reduced oral leukocyte activity, and
 - ✓ Proliferation of anaerobic, fusospirochetal microorganism.

EFFECTS OF SMOKING ON PERIODONTAL THERAPY

Table 14.4: Effects of smoking on periodontal therapies

Therapy	Effect of smoking
Nonsurgical	• Decreased clinical response to scaling and root planing • Decreased reduction in pocket depth • Decreased gain in clinical attachment level • Decreased negative impact of smoking with increased level of plaque control
Surgery and implants	• Decreased pocket depth reduction postsurgery • Increased deterioration of furcation postsurgery • Decreased gain in clinical attachment level, decreased bone fill, increased recession and increased membrane exposure following guided tissue regeneration (GTR) • Decreased pocket depth reduction after demineralized freeze dried bone allograft (DFDBA) • Decreased pocket depth reduction and gain in clinical attachment level after open flap debridement • Conflicting data on the impact of smoking on implant success • Smoking cessation should be recommended prior to implant
Maintenance	• Increased pocket depth during maintenance • Decreased gain in clinical attachment level • Recurrent (refractory) disease • Increased recurrent/refractory disease in smokers • Increased need for retreatment in smokers • Increased need for antibiotics in smokers to control the negative effect of periodontal infection on surgical outcome • Increased tooth loss in smokers after surgical therapy

SMOKING AND ORAL HEALTH

How can tobacco use effect my oral health?

England



BAD BREATH

The chemicals in cigarette smoke can linger in the mouth and lungs for hours causing the characteristic smoker's breath.



STAINING

The nicotine and tar in tobacco can make teeth go yellow in a very short time. Some of this staining can be permanent.



GUM DISEASE

Smoking can increase the number of bacteria in the mouth that cause gum disease. In severe cases this can cause the teeth to fall out.



CANCER

Two thirds of mouth cancer cases are linked to smoking.
Second-hand smoke increases the risk for those who have never smoked.
Ex-smokers reduce their risk by a third.



SMOKING CESSATION

- Patient counselling ; important 5As
 - ✓ **ASK** about current and past smoking status
 - ✓ **ADVISE** all smokers to quit, be informative , honest and helpful but not judgemental
 - ✓ **ASSESS** smokers' current stage; patients interest and readiness to attempt smoking cessation
 - ✓ **ASSIST** the patient in the preparation stage including nicotine replacement therapy(NRT)
 - ✓ **ARRANGE** for follow-up; referral to professional

CONT'D

5 A'S OF TOBACCO CESSATION



CONT'D

TRANSTHEORETICAL MODEL OF SMOKING CESSATION



Precontemplation stage	Contemplation stage	Preparation stage	Action stage	Maintenance stage
Patient is not aware that the habit is not the cause; no intention of quitting - increase the awareness and guide them into contemplation stage	Aware of the fact that smoking is the problem - expresses intention to stop within 6 months - reinforce and encourage the patient	Patient intends to stop the habit within less than 1 month - previously attempted to quit but without success - Adjunctive therapies - for cessation	Patient stops the habit and maintains early abstinence - patient made to control withdrawal symptoms; or chances of relapse	Patient has stopped the habit for 6 continuous months; continued efforts to be made to prevent relapse

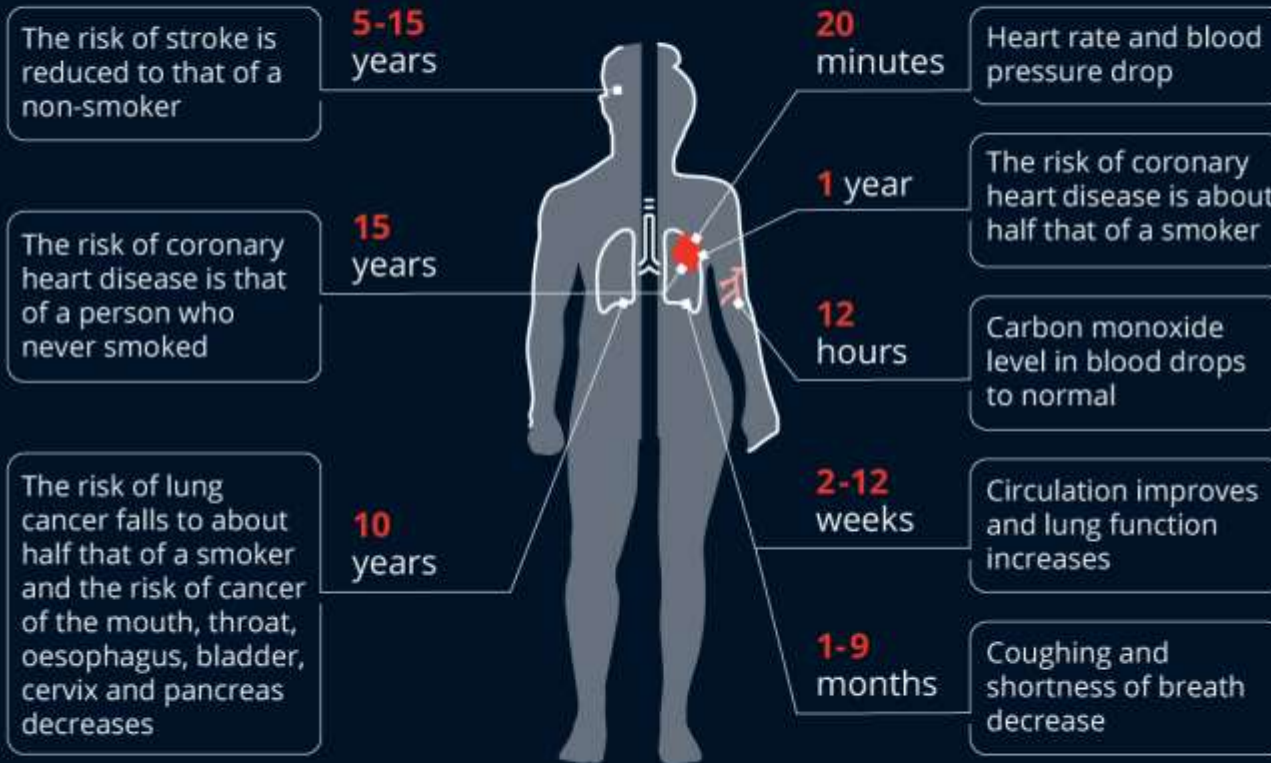
NRT

- Replace nicotine and gradually lower patients' dependency on smoking
 - ✓ **Nicotine patch**
 - ✓ **Nicotine gum**
 - ✓ **Nicotine nasal spray**
 - ✓ **Nicotine oral inhaler**

The Effects of Quitting Smoking

Health improvements that take place after quitting smoking, by time required

Time after quitting smoking:



Source: World Health Organization



ANY QUESTIONS???